

EXERCISE 1

The log on the right corresponds to a schedule of three *serially executing* transactions T_1 , T_2 and T_3 where a system failure occurs at the point indicated. Assume that the *immediate database modification protocol* is used.

a) Which transactions are undone?

Undo: T_3

b) Which transactions are redone?

Redo: T_1 T_2

Log File
< T_1 start>
< T_1 , B, 15, 12>
< T_1 commit>
< T_2 start>
< T_2 , B, 12, 18>
< T_2 , D, 25, 26>
< T_2 commit>
< T_3 start>
< T_3 , D, 20, 25>

← system failure

Scan the log and
undo if < T_i start> but no < T_i commit>
redo if < T_i start> and < T_i commit>

EXERCISE 2

The log on the right corresponds to a schedule of three concurrently executing transactions T_1 , T_2 and T_3 where a system failure occurs at the point indicated. Assume that the immediate database modification protocol with checkpointing is used.

a) Which transactions are undone?

Undo: T_2

b) Which transactions are redone?

Redo: T_3 T_1

Scan the log backward and
 if find $\langle T_i \text{ commit} \rangle$ add to redo-list
 if find $\langle T_i \text{ start} \rangle$ add to undo-list if not in redo-list
 for every T_i in $\langle \text{checkpoint}, \{T_i, \dots\} \rangle$
 if T_i not in redo-list add to undo-list

Log File	
$\langle T_1 \text{ start} \rangle$	
$\langle T_1, A, 2, 5 \rangle$	
$\langle T_1, B, 1, 2 \rangle$	
$\langle T_2 \text{ start} \rangle$	
$\langle T_1, C, 2, 8 \rangle$	
$\langle T_2, B, 2, 5 \rangle$	
$\langle \text{checkpoint}, \{T_1, T_2\} \rangle$	← add T_2 to undo-list
$\langle T_1 \text{ commit} \rangle$	← add to redo-list
$\langle T_2, B, 5, 6 \rangle$	
$\langle T_3 \text{ start} \rangle$	✓ on redo-list
$\langle T_3, A, 5, 3 \rangle$	
$\langle T_3 \text{ commit} \rangle$	← add to redo-list
$\langle T_2, C, 8, 2 \rangle$	
$\langle T_2, A, 3, 5 \rangle$	
	← system failure

EXERCISE 3

The log on the right corresponds to a schedule of four serially executing transactions T_1 , T_2 , T_3 and T_4 where a system failure occurs at the point indicated. Assume that the immediate database modification protocol with checkpointing is used.

a) Which transactions are undone?

Undo: T_4

b) Which transactions are redone?

Redo: T_1 T_2 T_3

Log File
< T_1 start>
< T_1 , A, 1, 2>
< T_1 , B, 5, 12>
<checkpoint>
< T_1 commit>
< T_2 start>
< T_2 , B, 12, 18>
< T_2 commit>
< T_3 start>
< T_3 , D, 25, 26>
< T_3 commit>
< T_4 start>
< T_4 , A, 20, 25>

← system failure

Scan log backward to first < T_i start> before the checkpoint.
Scan log forward and
undo if < T_i start> but no < T_i commit>
redo if < T_i start> and < T_i commit>

EXERCISE 4

The log on the right corresponds to a schedule of three concurrently executing transactions T_1 , T_2 and T_3 where a system failure occurs at the point indicated. Assume that the immediate database modification protocol with checkpointing is used.

a) Which transactions are undone?

Undo: T_1 T_3

b) Which transactions are redone?

Redo: T_2

Log File	
< T_3 start>	
< T_3 , B, 15, 12>	
< T_2 start>	
<checkpoint { T_2 , T_3 }>	← add T_3 to undo-list
< T_1 start>	← add T_1 to undo-list
< T_2 , B, 12, 18>	
< T_1 , D, 20, 25>	
< T_2 commit>	← add T_2 to redo-list
< T_3 , A, 1, 2>	
< T_1 , D, 25, 26>	
	← system failure

Scan the log backward and

if find < T_i commit> **add to** redo-list

if find < T_i start> **add to** undo-list **if not in** redo-list

for every T_i **in** <checkpoint, { T_i , ...}>

if T_i **not in** redo-list **add to** undo-list